Assignment-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

• Prompt AI to convert a Python class with attributes and methods into equivalent Java code.

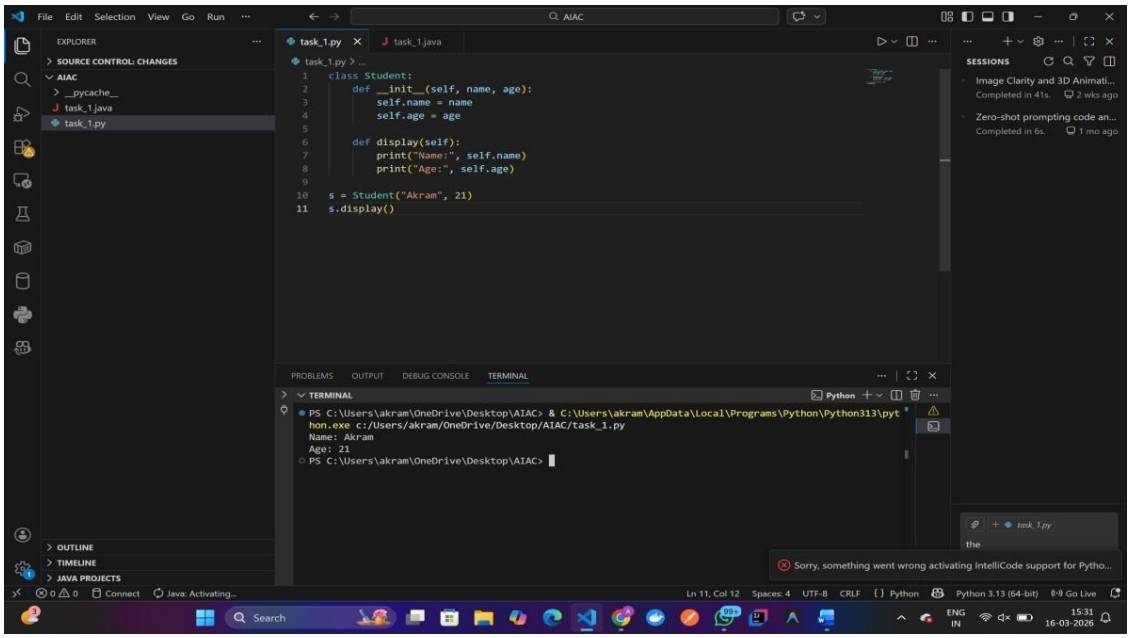
• Verify that constructors, method signatures, and data types are properly handled.

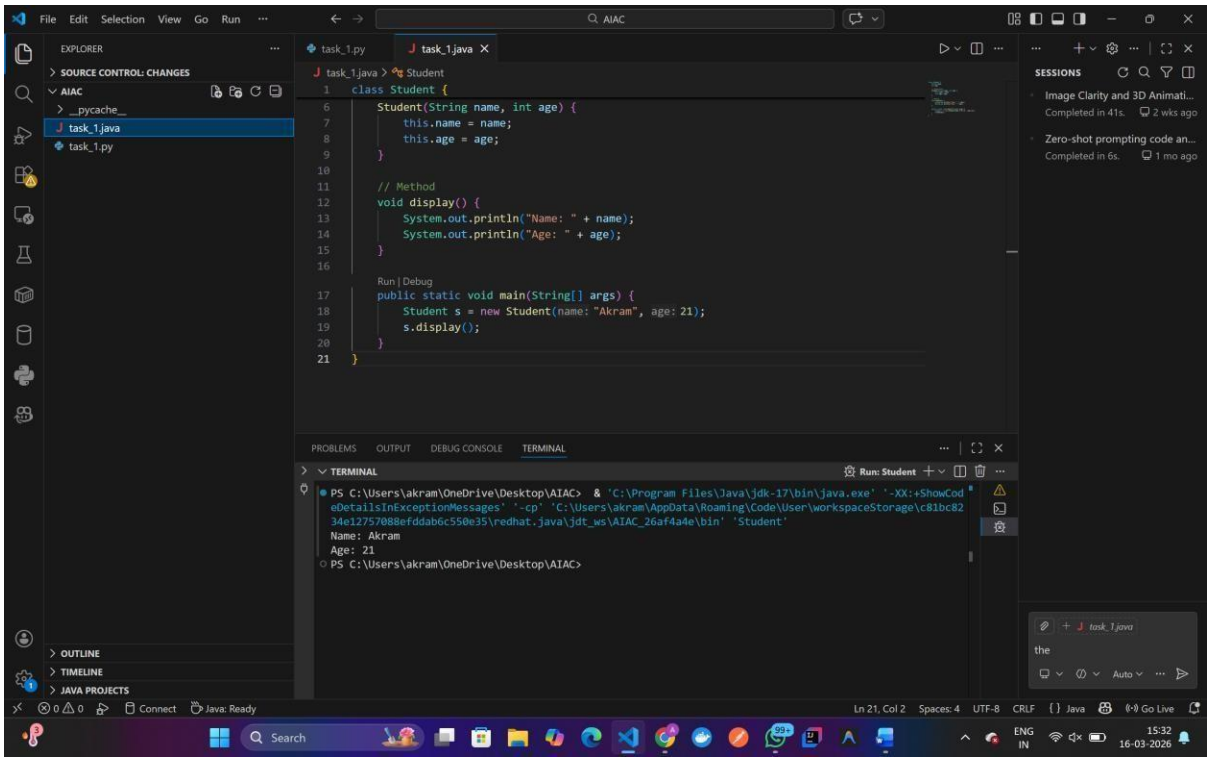
• Run both versions (Python & Java) to compare expected outputs.

Expected Output:

• Equivalent working Java class translated from Python.

CODE:





Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

• Ask AI to translate a recursive C++ factorial function into

Python.

• Validate correctness by testing with sample inputs.

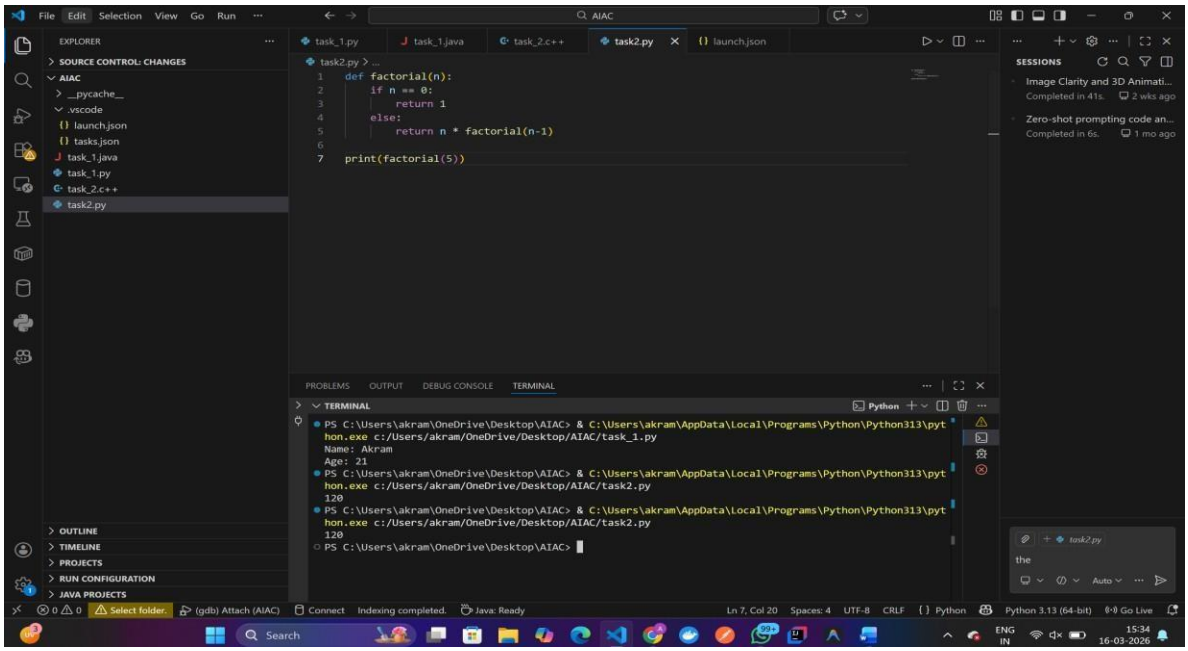
• Ensure Python version follows Pythonic conventions.

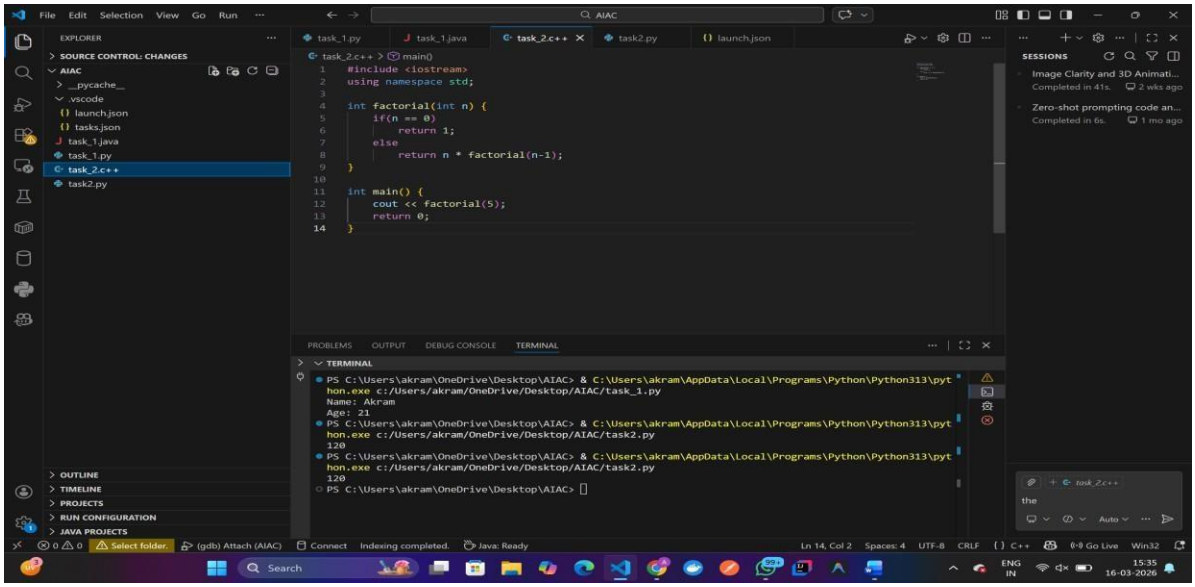
Expected Output:

• A correct Python implementation equivalent to the original

C++ factorial function.

CODE:





Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in

Python.

Instructions:

• Provide AI with a SQL query (e.g., SELECT name, salary

FROM employees WHERE salary > 50000).

• Ask AI to generate equivalent Pandas code.

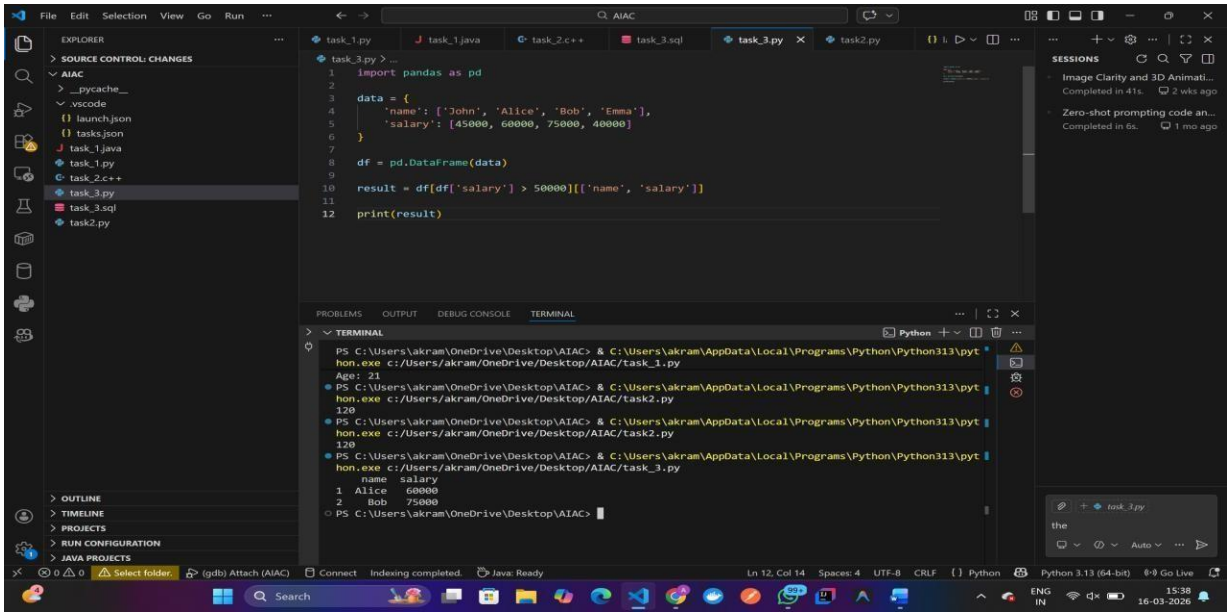
• Test the generated code on a sample DataFrame.

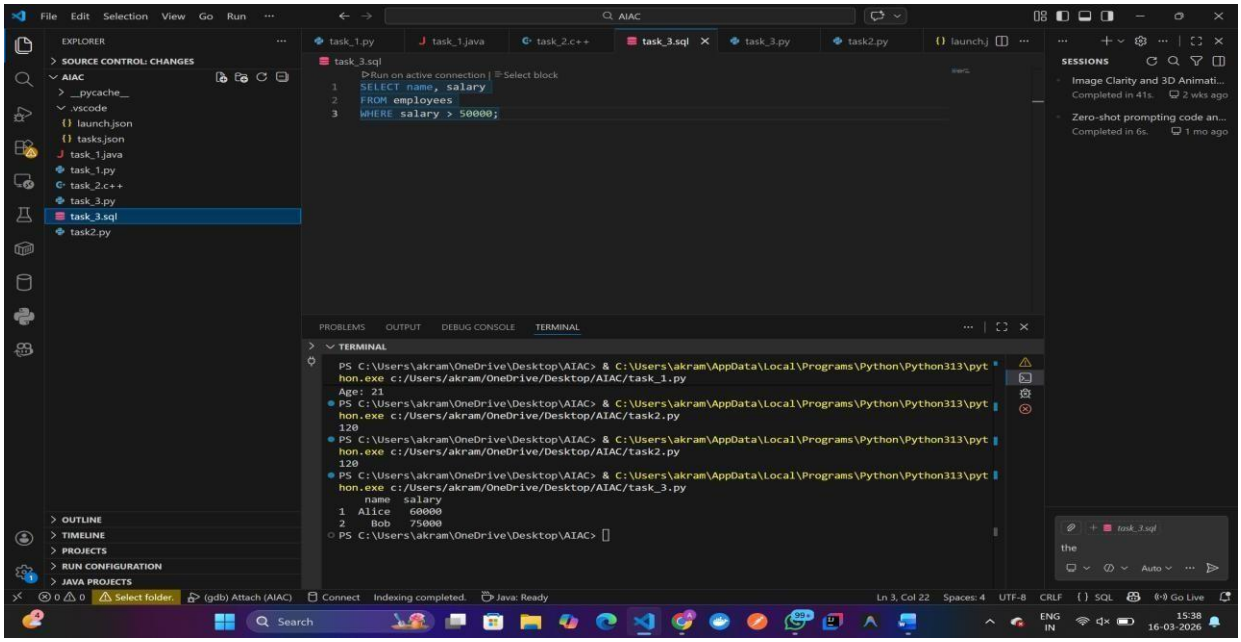
Expected Output:

• Equivalent Pandas query that retrieves the correct subset of

data.

CODE:





Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in

multiple languages. Example:

• Converting a Python web scraping snippet into JavaScript for browser automation.

• Translating a Java method into C# for enterprise applications.

Instructions:

• Prompt AI to translate code between two chosen languages.

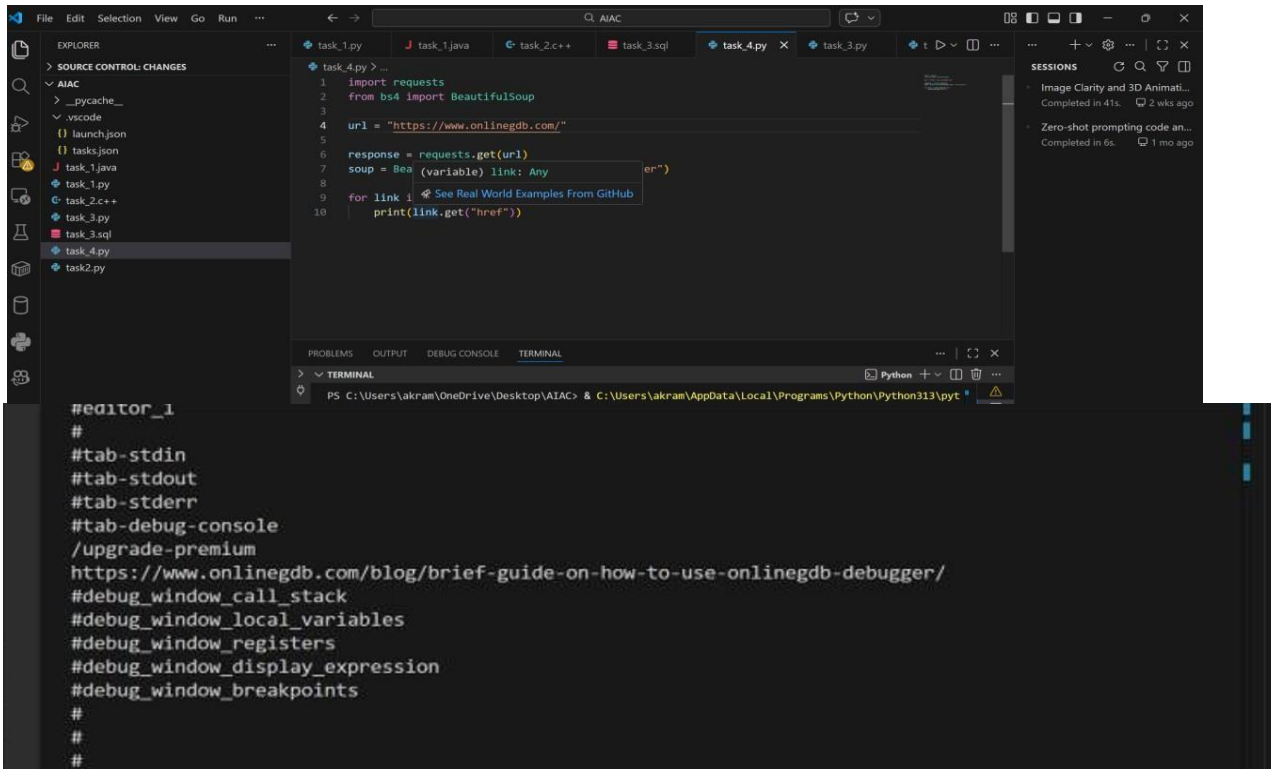
• Test both implementations to ensure they produce the same results.

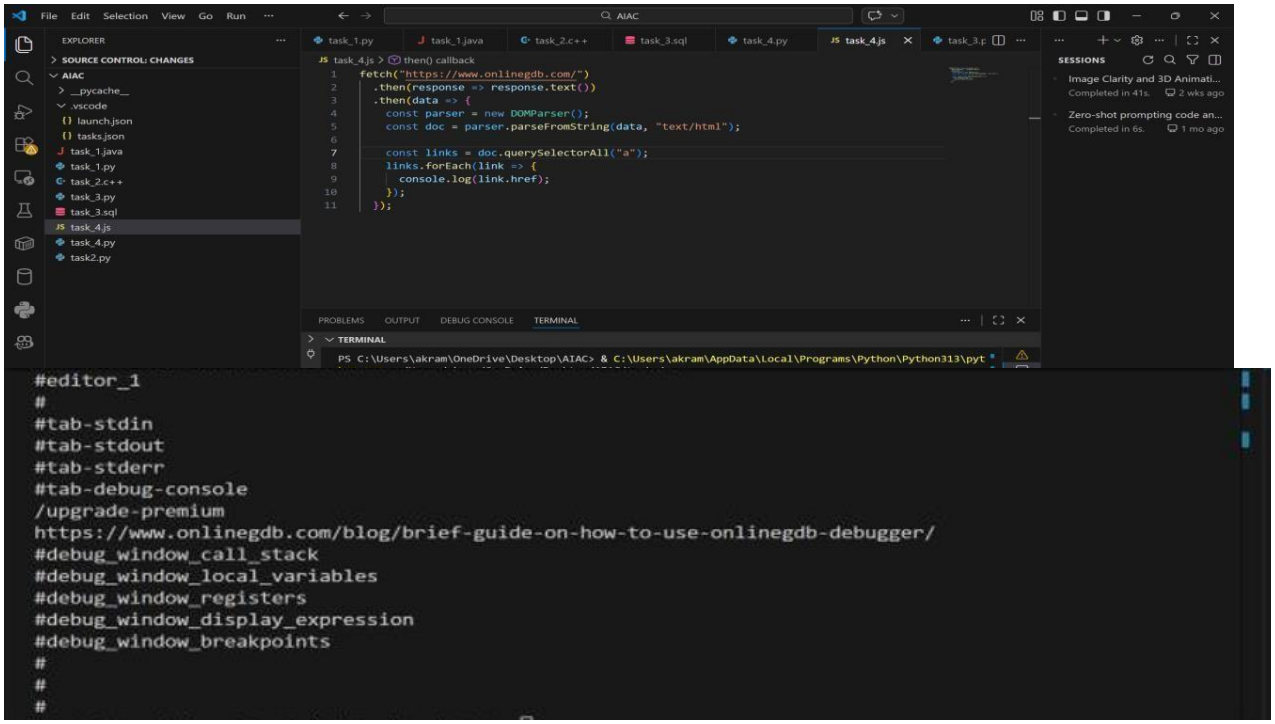
• Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

• Working equivalent code snippets in at least two different programming languages

CODE:





Explanation: In this assignment, we used AI to convert code from one programming language to another. We converted Python to Java, C++ to Python, SQL to Pandas, and also worked on a real-time example.
We checked that all converted programs give correct output. This helped us understand differences between languages and how AI can make coding easier and faster.